확률·랜덤프로세스 기말대비 6/13 (D7) · 2차 — 인터리브 복습 L07–L09

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- L07(공분산/변환)·L08(상관/반복기댓값/전분산)·L09(CLT) 공식 인출.

문항 목차 (총 10문항)

Problem 31
Consider two random variables X and Y with joint PMF given in Table 5.6.

Table 5.6: Joint PMF of X and Y in Problem 31

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

Find $Cov(X, Y)$ and $\rho(X, Y)$.

— 풀이 —

Problem 33

Let X and Y be two random variables. Suppose that $\sigma_X^2 = 4$, and $\sigma_Y^2 = 9$. If we know that the two random variables $Z = 2X - Y$ and $W = X + Y$ are independent, find $Cov(X, Y)$ and $\rho(X, Y)$.

— 풀이 —

5.8.3● For the random variables X and Y in Problem 5.2.2 find

(a) The expected value of $W = 2^{XY}$,

(b) The correlation, $r_{X,Y} = \mathrm{E}[XY]$,

(c) The covariance, $\mathrm{Cov}[X, Y]$,

(d) The correlation coefficient, $\rho_{X,Y}$,

(e) The variance of $X + Y$, $\mathrm{Var}[X + Y]$.

(Refer to the results of Problem 5.3.2 to answer some of these questions.)

— 풀이 —

6.3.1 X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/3 + 1/3 & -1 \le x < 0, \\ x/3 + 2/3 & 0 \le x < 1, \\ 1 & 1 \le x. \end{cases}$$

$Y = g(X)$ where

$$g(X) = \begin{cases} 0 & X < 0, \\ 100 & X \ge 0. \end{cases}$$

(a) What is $F_Y(y)$?

(b) What is $f_Y(y)$?

(c) What is $\mathrm{E}[Y]$?

— 풀이 —

7.5.7 Over the circle $X^2 + Y^2 \le r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \le r^2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is $f_{Y|X}(y|x)$?

(b) What is $\mathrm{E}[Y|X = x]$?

— 풀이 —

7.6.4◆ Let X_1 and X_2 have a bivariate Gaussian PDF with correlation coefficient ρ_{12} such that each X_i is a Gaussian (μ_i, σ_i) random variable. Show that $Y = X_1 X_2$ has variance

$$\begin{aligned}\operatorname{Var}[Y] = \sigma_1^2\sigma_2^2(1+\rho_{12}^2)\\ + \sigma_1^2\mu_2^2 + \mu_1^2\sigma_2^2 - \mu_1^2\mu_2^2.\end{aligned}$$

Hints: Look ahead to Problem 9.2.4 and also use the iterated expectation to find

$$\mathrm{E}\left[X_1^2X_2^2\right] = \mathrm{E}\left[\mathrm{E}\left[X_1^2X_2^2|X_2\right]\right].$$

— 풀이 —

Problem 5

The amount of time needed for a certain machine to process a job is a random variable with mean $EX_i = 10$ minutes and $\mathrm{Var}(X_i) = 2$ minutes^2. The times needed for different jobs are independent from each other. Find the probability that the machine processes less than or equal to 40 jobs in 7 hours.

— 풀이 —

10.4.1 ● When X is Gaussian, verify Equation (10.24), which states that the sample mean is within one standard error of the expected value with probability 0.68.

— 풀이 —

2.1.1 ● Suppose you flip a coin twice. On any flip, the coin comes up heads with probability 1/4. Use H_i and T_i to denote the result of flip i.

(a) What is the probability, $P[H_1|H_2]$, that the first flip is heads given that the second flip is heads?

(b) What is the probability that the first flip is heads and the second flip is tails?

— 풀이 —

5.2.2 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} c|x+y| & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is P[$Y < X$]?

(c) What is P[$Y > X$]?

(d) What is P[$Y = X$]?

(e) What is P[$X < 1$]?

— 풀이 —